

Artificial Intelligence & Systems Engineering

Chapter 1: Introduction to Modern Systems Engineering

In modern software development, specifically in engineering artificial intelligence systems, modularity serves as the foundational cornerstone that dictates long-term reliability. Modular component design works by isolating system complexity into independent, single-responsibility modules.

The primary goals of this modular approach are:

1. Preventing functional regressions when performing codebase updates.
2. Facilitating isolated unit testing of individual services.
3. Accelerating parallel component iterations across development teams.

By isolating complex parameters into controlled boundaries, engineers can minimize cross-module dependencies and maximize the operational stability of the overall system. This approach is proven to reduce development time and mitigate risks of systematic errors as applications scale.

Chapter 2: Vector Databases & pgvector Indexing

High-performance semantic retrieval requires dedicated storage infrastructures optimized for high-dimensional vector representations. Vector databases emerge as the optimal solution to store, index, and rapidly query mathematical representations of text or study materials.

Within the PostgreSQL ecosystem, enabling the pgvector extension allows storing vector embeddings directly in database tables. pgvector supports index generation using specialized structures such as IVFFlat (Inverted File with Flat Compression) or HNSW (Hierarchical Navigable Small World).

These indexing structures enable similarity searches to execute within milliseconds. The similarity is calculated using the Cosine Distance mathematical formula, which measures the angular difference between two high-dimensional vectors:

$$\text{Cosine Similarity} = (a \cdot b) / (\|a\| * \|b\|)$$

By leveraging pgvector, we can easily locate the most relevant paragraphs from thousands of documents simply by comparing their vector similarities.

Chapter 3: Analyzing & Solving Overfitting

One of the greatest challenges in training machine learning models is the phenomenon of overfitting. A model that achieves 99% training accuracy but falls to 60% on validation is demonstrating overfitting. This is a state where a machine learning model memorizes specific details and noise in the training dataset rather than learning generalizable, universal patterns.

Core characteristics of an overfitted model include:

1. High training accuracy (near-perfect performance).
2. Poor and highly unstable test or validation accuracy.
3. High variance in response to small perturbations in input features.

To mitigate overfitting, machine learning practitioners are recommended to:

- Apply L1 (Lasso) or L2 (Ridge) regularization to penalize excessively large weights.
- Simplify the neural network architecture by reducing layers or node density.
- Expand and diversify the training dataset through data augmentation or collecting new samples.
- Employ early stopping techniques to halt training before the model starts memorizing noise.

Chapter 4: RAG Architectures & Token Efficiency

Retrieval-Augmented Generation (RAG) is a highly efficient technique that combines search retrieval with generative LLMs. Sending the entire content of a PDF file to a large language model (LLM) like OpenAI is highly inefficient, quickly consuming context token limits and incurring expensive operational costs.

The RAG architecture solves this through a structured workflow:

1. The PDF document is split into smaller segments using a sentence-boundary parser (semantic chunking).
2. Each segment is converted to a vector embedding (e.g., 1536 dimensions).
3. When the user asks a question, the system queries pgvector for the top 3 to 5 most similar segments.
4. Only those retrieved context segments are sent to OpenAI as context to generate an answer.

This selective approach dramatically minimizes OpenAI token consumption, speeds up system query response times, and guarantees highly accurate answers grounded in original source material without information hallucination.